

Pierre Kawak, Ph.D.

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Education & Training

University of Wisconsin, Madison (UW-Madison)	Postdoctoral Scholarship	2026 –
<i>Advisor: Ying Li</i>		
University of South Florida (USF)	Postdoctoral Scholarship	2022 – 2025
<i>Advisor: David S. Simmons</i>		
Brigham Young University (BYU)	Ph.D. Chemical Engineering	2017 – 2022
<i>Advisor: Douglas R. Tree</i>		
Dissertation: Simulation of Crystal Nucleation in a Polymer Melt		
American University of Sharjah (AUS)	M.S. Chemical Engineering	2015 – 2017
<i>Advisor: Ghaleb A. Hussein</i>		
Dissertation: Ultrasound Triggered Release of Estrone-Targeted Liposomes		
American University of Sharjah (AUS)	B.S. Chemical Engineering	2010 – 2015
Minor Economics		

Publications

Peer-Reviewed Journal Articles

- [6] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Glassy interphases reinforce elastomeric nanocomposites by enhancing volume expansion under strain”. *Proceedings of the National Academy of Sciences* 123 (2026). DOI: [10.1073/pnas.2528108123](https://doi.org/10.1073/pnas.2528108123).
- [5] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Origin of Heating-Induced Softening and Enthalpic Reinforcement in Elastomeric Nanocomposites”. *ACS Macro Letters* 14 (2025). DOI: [10.1021/acsmacrolett.5c00442](https://doi.org/10.1021/acsmacrolett.5c00442).
- [4] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Central Role of Filler-Polymer Interplay in Nonlinear Reinforcement of Elastomeric Nanocomposites”. *Macromolecules* 57 (2024). DOI: [10.1021/acs.macromol.4c00489](https://doi.org/10.1021/acs.macromol.4c00489).
- [3] **Pierre Kawak**, Christopher Akiki, and Douglas R. Tree. “Effect of local chain stiffness on oligomer crystallization from a melt”. *Physical Review Materials* 8 (2024), p. 075606. DOI: [10.1103/PhysRevMaterials.8.075606](https://doi.org/10.1103/PhysRevMaterials.8.075606).
- [2] **Pierre Kawak**, Dakota S. Banks, and Douglas R. Tree. “Semiflexible oligomers crystallize via a cooperative phase transition”. *Journal of Chemical Physics* 155 (2021), p. 214902. DOI: [10.1063/5.0067788](https://doi.org/10.1063/5.0067788).
- [1] Najla M. Salkho, Vinod Paul, **Pierre Kawak**, Rute F. Vitor, Ana M. Martins, Mohammad Al Sayah, and Ghaleb A. Hussein. “Ultrasonically controlled estrone-modified liposomes for estrogen-positive breast cancer therapy”. *Artificial Cells, Nanomedicine, and Biotechnology* 46 (2018), pp. 462–472. DOI: [10.1080/21691401.2018.1459634](https://doi.org/10.1080/21691401.2018.1459634).

Preprints

- [3] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Fracture of Elastomeric Nanocomposites Due to Filler Pillar Formation” (in preparation).

- [2] Douglas R. Tree and **Pierre Kawak**. “The Search for a Molecular-Level Understanding of Nucleation in Polymer Crystallization” (in preparation).
- [1] **Pierre Kawak**, William F. Drayer, Mark E. Mackura, and David S. Simmons. “Amorphous Molecular Dynamics Analysis Toolkit (AMDAT): A Software Package for Analysis of Simulations of Structure and Dynamics in Supercooled Liquids and Complex Fluids” (2026). DOI: [10.48550/arXiv.2602.05865](https://doi.org/10.48550/arXiv.2602.05865).

Software

- [2] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Filled Elastomer Simulation Package (FESP)*. Version 1.0.0. 2026. DOI: [10.5281/zenodo.19362854](https://doi.org/10.5281/zenodo.19362854).
- [1] David Simmons, **Pierre Kawak**, William Drayer, and Mark Mackura. *Amorphous Molecular Dynamics Analysis Toolkit (AMDAT)*. Version 1.0.1. 2025. DOI: [10.5281/zenodo.17417166](https://doi.org/10.5281/zenodo.17417166).

Datasets

- [6] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Dataset for Glassy interphases reinforce elastomeric nanocomposites by enhancing volume expansion under strain*. Dataset. 2026. DOI: [10.17632/dg9jxp8sj6](https://doi.org/10.17632/dg9jxp8sj6).
- [5] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Dataset for Origin of Heating-Induced Softening and Enthalpic Reinforcement in Elastomeric Nanocomposites - Young's modulus and Poisson's ratio versus temperature data from uniaxial extension molecular dynamics simulations of model elastomers*. Dataset. 2026. DOI: [10.17632/c4bnns89g2](https://doi.org/10.17632/c4bnns89g2).
- [4] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Dataset for Origin of Heating-Induced Softening and Enthalpic Reinforcement in Elastomeric Nanocomposites - Bulk modulus versus temperature data from equilibrium molecular dynamics simulations of a model neat elastomer*. Dataset. 2026. DOI: [10.17632/rfrb249t77](https://doi.org/10.17632/rfrb249t77).
- [3] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Dataset for Origin of Heating-Induced Softening and Enthalpic Reinforcement in Elastomeric Nanocomposites - Figure 1 - Render of a filled elastomer configuration*. Dataset. 2026. DOI: [10.17632/vrd38jc7hp](https://doi.org/10.17632/vrd38jc7hp).
- [2] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Dataset for Origin of Heating-Induced Softening and Enthalpic Reinforcement in Elastomeric Nanocomposites - Figure 2 - Relaxation time versus temperature*. Dataset. 2026. DOI: [10.17632/djm3kdvxwp](https://doi.org/10.17632/djm3kdvxwp).
- [1] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Dataset for Origin of Heating-Induced Softening and Enthalpic Reinforcement in Elastomeric Nanocomposites - Figure 3 - Young's modulus, bulk modulus, and Poisson's ratio versus temperature with validation of Poisson's-ratio-mismatch theory*. Dataset. 2026. DOI: [10.17632/4bs9sxxb5](https://doi.org/10.17632/4bs9sxxb5).

Awards & Fellowships

Research Grants & Fellowships

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| National Postdoctoral Association (NPA) IMPACT Fellowship | 2023 – 2024 |
| One of six selected out of 100 applicants nationwide for funding & mentorship of proposed project. | |
| National Science Foundation (NSF) CoPI Discover ACCESS Compute Resource Grant | Nov. 2023 |
| Awarded NSF funding for access to high performance computing resources. | |
| American Physical Society (APS) Career Mentoring Fellowship | 2023 |
| Received mentorship training, administered career talk at USF, & judged young trainee talks. | |

Program Acceptance & Participation

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| Torrey Pines Foundations of Leadership Development Program Participant | 2024 – 2025 |
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NSF & SACNAS Grant Writing & Peer Review Workshop Attendance	Aug. 2023
Future Faculty Workshop Diverse Leaders for the Future Workshop Attendance	June 2023
Out in Science, Tech., Engineering, & Maths Professional Development Summit Participant	Mar. 2021
UCSD SDSC High Performance Computing Summer Institute Attendee	Jul. 2018

Conference Awards

Outstanding Poster Award at Gordon Research Conference on Polymer Physics	July 2024
USF Annual Postdoctoral Research Symposium Best Poster Award \$200	Mar. 2023
APS Forum on International Physics Distinguished Student Award	Fall 2022

Excellence, Leadership & Service Awards

AUS College of Engineering Hall of Fame Inductee	2023
BYU Chemical Engineering Department Graduate Student of the Month	Sept. 2022
BYU University Accessibility Center Banquet Scholarship Award \$1,500	Fall 2021
BYU Graduate Student Society Professional Presentation Award \$500	Fall 2021
BYU Chemical Engineering Department Travel Award	Fall 2021
Delta Alpha Pi (DAPI) International Honor Society Inductee	2021
AUS Biomedical Engineering Symposium Best Overall Talk Award \$700	Fall 2016
AUS 3× Dean's List for Academic Excellence	2013 – 2014

Research Mentorship Experience

Amanda Sharrer	<i>Ph.D. Candidate, USF</i>	2024-2025
– Onboarded and provided ongoing technical guidance for a project on molecular sequence selection for additive compatibilizers in mixed waste plastics. – Assisted in building molecular configurations and simulation protocols (equilibration/production, validation). – Supported analysis and visualization workflows using Python and OVITO to probe interfacial phenomena.		
Luis Zepeda	<i>Ph.D. Candidate, USF</i>	2024-2025
– Provided technical guidance for simulations of thin polymer films near supported and free surfaces. – Helped establish reliable simulation protocols to study structure and dynamics near interfaces. – Supported Python post-processing and visualization to extract interfacial trends and communicate results clearly.		
Harshad Bhapkar	<i>Ph.D. Candidate, USF</i>	2022-2025
– Directly supervised day-to-day research activities focused on mechanical response of polymer nanocomposites. – Provided hands-on mentorship spanning simulation planning, troubleshooting, analysis strategy, and interpretation of mechanical/structural metrics. – Served as a primary technical point-of-contact to maintain momentum, prioritize tasks, and ensure method rigor.		
Makayla Branham	<i>Ph.D. Candidate, USF</i>	2023-2025
– Directly mentored and collaborated on research studying copolymer sequence effects on glass transition behavior. – Guided analysis, planning and interpretation to connect sequence-level descriptors with relaxation metrics.		
Bao Ma	<i>Ph.D. Candidate, USF</i>	2023-2025
– Onboarded and provided technical guidance for thin-film polymer simulations near interfaces. – Supported development of simulation protocols and Python analysis for interfacial structure and dynamics.		
Alyna Williams	<i>Undergraduate Researcher, USF</i>	2024-2025
– Mentored end-to-end research execution: scientific computing, HPC usage, simulation workflows, and polymer physics literature grounding.		

- Co-developed generalized relaxation characterization pipeline across diverse molecular systems.
- Provided career mentorship and wrote letters of reference in support of professional opportunities.

Tianna Virgo *Undergraduate Researcher, USF* 2024

- Onboarded into simulation-based polymer physics research with structured guidance in scientific computing workflows, high-performance computing procedures, and best practices.

Dr. Peijing Yue *Ph.D. Candidate, USF* 2022-2024

- Provided sustained professional development/guidance during the Ph.D. to industry transition.
- Helped translate research strengths into industry-ready narratives and practical application materials.

Annelise Curtin *M.S. Student, USF* 2024

- Provided career guidance for graduate-school prep; placed as Ph.D. at the University of Minnesota.

Austin Hartley *Undergraduate Researcher, USF* 2023

- Provided career guidance for graduate-school prep; placed as Ph.D. at the Penn State University.

Dr. William F. Drayer *Ph.D. Candidate, USF* 2022-2023

- Provided professional development and career guidance during transition from Ph.D. to postdoc.

Christopher Akiki *Undergraduate Researcher, BYU* 2020-2022

- Co-authored a peer-reviewed manuscript on sensitivity of polymer crystallization to force-field choices.
- Collaborated on development of the Order-Parameter Wang-Landau (OPWL) simulation engine in C++, a generalized rare-event sampler for free energies and densities of states.
- Provided technical guidance on literature review and methodological approaches to polyethylene crystallization.
- Provided career guidance for graduate-school prep; placed as Ph.D. at University of Minnesota.

Dr. Dakota S. Banks *Undergraduate Researcher, BYU* 2018-2019

- Co-authored a peer-reviewed manuscript on minimum free energy path for polymer crystal nucleation.
- Collaborated on the design and implementation of a GPU-accelerated Monte Carlo Polymer Crystallization (MCPC) simulation engine in C++/CUDA.
- Provided career guidance for graduate-school prep; placed as Ph.D. at Brigham Young University.

Beverly S. Delgado *Undergraduate Researcher, BYU* 2019

- Supported onboarding into molecular simulations and polymer physics with literature and resources.

Andrew S. Gibson *Undergraduate Researcher, BYU* 2018

- Collaborated on development of the Monte Carlo Polymer Crystallization (MCPC) simulation engine in C++/CUDA and provided onboarding training in simulation workflows.

Paul Kawak *M.S. Student, AUS* 2015-2017

- Provided onboarding/mentorship on a project developing albumin-targeted smart nanocarriers for cancer treatment (synthesis, validation, and release studies probing ultrasonic-triggered response).

Teaching Experience

Graduate Teaching Assistant	Thermodynamics	Winter 2021
<i>Brigham Young University</i>	Separations Engineering	Fall 2021
	Heat & Mass Transfer	2018 – 2021 (3x)
	Process Dynamics & Control	Fall 2018
Volunteer Course Instructor; <i>University of the People</i>	College Algebra	Spring 2018
Graduate Instructor; <i>American University of Sharjah</i>	Principles of ChemE	2016 – 2017 (3x)

Graduate Teaching Assistant <i>American University of Sharjah</i>	Corrosion Lab ChemE Lab I Graduate Desalination Wastewater Treatment	2016 – 2017 (2x) 2015 – 2016 (2x) Spring 2015 Spring 2015
Undergraduate Teaching Assistant <i>American University of Sharjah</i>	Mass Transfer Kinetics Thermodynamics	2014 – 2015 (3x) Fall 2014 Spring 2014
Private Tutor	Maths, Engineering, Business, etc.	2008 – now

Community & Service

Peer Review

American Chemical Society Macromolecules	9× Reviews
American Chemical Society Journal of Chemical Information and Modeling	2× Reviews
American Chemical Society Petroleum Research Fund (PRF)	2× Review
Freiburg Institute for Advanced Studies Early Career Fellowship Programme	1× Review
Wiley Journal of Polymer Science	1× Review

Professional Society & Conference Leadership

Co-Chair, Gordon Research Seminar on Polymer Physics	2026
Organized technical sessions and networking; fundraised for the seminar.	
<i>American Physical Society (APS)</i>	
Session Chair & Organizer: “Polymer Structure & Dynamics across Length & Time Scales”	2024, 2025, 2026, 2027
Session Chair & Organizer: “Tuning Polymer Properties Through Macromolecular Architecture & Interactions”	2025, 2026, 2027
Member of Committee on International Freedom of Scientists	2025-2026
Member of Division of Polymer Physics (DPOLY) Membership Committee	2024-2025
Organizer & Winner of Inaugural DPOLY T-Shirt Design Competition	Mar 2024
Ranked Undergraduate Talks & Posters	2022 – 2023
Session Chair “ Polymers & Composites for Energy Storage & Conversion I ”	Mar 2023
<i>Early Career Researchers in Polymer Physics</i>	
Administrator of 550 member Slack channel for collaboration & networking	2022 – present
Organizer, Virtual Polymer Physics Symposium (150+ attendees)	Aug. 2023
Cofounder & Organizer of Self-Development Seminar series	2022 – present
<i>Out in Science Technology Engineering & Mathematics (oSTEM), Inc.</i>	
Represented oSTEM at MAA MathFest outreach tables.	Aug. 2023
Annual Conference Volunteer & Organizer	Nov. 2022
Annual Conference Merchandise Team Organizer	Nov. 2022

Departmental & Institutional Service

<i>UW–Madison Postdoctoral Association</i>	
Executive Committee Treasurer	2026 – present
Fundraised for the 2026 Postdoctoral Research Symposium.	2026

University of South Florida (USF) Postdoctoral Scholar Association (PSA)

Founded & Chaired PSA Executive Committee at USF serving 200 postdocs	2023 – present
Organized Inaugural ELEVATE Talk Series	2024 – 2025
Led Postdoc Highlight Interviews, Socials, & Orientations	2023 – present

BYU Chemical Engineering Graduate Student Council (GSC)

President & Cofounder	2018 – 2021
Prepared onboarding materials and resources for new graduate students.	
Created graduate-student health and well-being surveys, including a financial-health survey administered in fall 2021.	
Organizer of Department Recruitment Poster Event	2019, 2020, 2021
Department BBQ Social Organizer	2018 – 2021
Social Media Accounts Manager	Fall 2021

Outreach, Inclusion & Public Engagement

Advisory Board Member, Pulse Physics	2026 – present
High-school physics club network.	

Presidential AI Challenge — Volunteer Judge & Mentor	2025
Posted machine-learning project guidance for parents and students across the U.S.; hosted office hours and judged inaugural-session entries.	

Volunteer Judge, State Science & Engineering Fair of Florida (SSEF)	2023, 2024
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Judged several Utah science fairs during Ph.D. training at BYU.

American Chemical Society (ACS)

Science Coaches	2023 – 2024
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Supported a Bradenton Christian School teacher with questions about her physics coursework by email and phone.

Bradenton Christian School, Florida — Full-Day Visit	Mar. 2024
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Discussed materials-science careers with around 100 grade 8–10 students.

Presented “Academia, Polymers, and the Ocean,” engaging students in a sustainability discussion that prompted ideas for addressing plastic waste.

American Physical Society (APS)

Member of National Mentoring Community	2022 – present
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[Physicists To-Go](#) Public Engagement Program Participant

Josephine C. Locke Elementary School, Chicago — Scholar Visit	Oct. 2023
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Spent a full day teaching and engaging with about 150 grade-8 students.

University of South Florida (USF)

Martinez Middle School — Great American Teach-In	Nov. 2024
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Delivered a talk on careers in physics.

Facilitated a summer workshop for high-school students on programming and scientific computing.	June 2023
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American Society for Engineering Education (ASEE)

Facilitator of Trans Allyship Safe Zone Ally Training Workshop	Mar. 2023
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Member of ASEE LGBTQ+ Advocacy in STEM Virtual Community of Practice	2022 – present
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Out in Science Technology Engineering & Mathematics (oSTEM), Inc.

Scholarship Coordinator; led applicant reviews.

2022 – 2023

Mentorship Program Volunteer

2021 – present

Technical Presentations

- [29] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Low-Strain Reinforcement as a Volume-Expansion Effect in Filled Elastomers with Glassy Interphases”. Global Physics Summit. American Physical Society. Denver, CO, 2026.
- [28] **Pierre Kawak**. “Molecular Simulations and Machine Learning for Sustainable Polymer Innovation”. Southeast Polymer Forum. University of Georgia, Athens. Athens, GA, 2025.
- [27] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Exploring the Role of Polymer-Filler Interactions in Modulating Elastomeric Reinforcement”. Global Physics Summit. American Physical Society. Anaheim, CA, 2025.
- [26] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Tuning Polymer-Filler Interactions to Modulate Elastomeric Reinforcement”. Rubber Division Spring Technical Meeting. American Chemical Society. Lake Buena Vista, FL, 2025.
- [25] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Tuning Reinforcement, Void Formation, and Fracture in Elastomeric Nanocomposites: Toward High-Performance, Sustainable Tire Materials”. Chemical, Biological, & Materials Engineering Department Seminar. University of South Florida. Tampa, FL, 2025.
- [24] **Pierre Kawak**. “Molecular Simulations for Greener Polymers: From Theory to Reality”. Annual Meeting. American Institute of Chemical Engineers. San Diego, CA, 2024.
- [23] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Contrasting Reinforcement Mechanisms in Elastomeric Nanocomposites”. Annual Meeting. American Institute of Chemical Engineers. San Diego, CA, 2024.
- [22] **Pierre Kawak**. “Filler-Filler Contacts Reinforce Filled Elastomers at High Strains”. GRC Polymer Physics. Gordon Research Conferences. South Hadley, MA, 2024.
- [21] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Filler-Filler Contacts Reinforce Filled Elastomers at High Strains”. Annual Postdoctoral Research Symposium. University of South Florida. Tampa, FL, 2024.
- [20] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Polymer-Filler Competition-Driven Reinforcement Beyond the Payne Effect in Elastomeric Nanocomposites”. March Meeting. American Physical Society. Minneapolis, MN, 2024.
- [19] Harshad Bhapkar, **Pierre Kawak**, and David S. Simmons. “Exploring the Effects of Nanoparticle Loading, Dispersion and Structure on the Stress Response of Elastomeric Nanocomposites”. March Meeting. American Physical Society. Minneapolis, MN, 2024.
- [18] **Pierre Kawak**, David S. Simmons, and Douglas R. Tree. “Rational Sustainable Polymer Materials Design Using Multiscale Simulation and Theory”. Annual Meeting. American Institute of Chemical Engineers. Orlando, FL, 2023.
- [17] **Pierre Kawak**, Makayla Branham, William F. Drayer, and David S. Simmons. “Tuning Polymer Dynamics Via Sequence Control”. Annual Meeting. American Institute of Chemical Engineers. Orlando, FL, 2023.
- [16] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Elucidating the Molecular Origins of Reinforcement in Filled Elastomers Via Spatial- and Species-Resolved Stresses from Molecular Dynamics Simulations”. Annual Meeting. American Institute of Chemical Engineers. Orlando, FL, 2023.

- [15] Harshad Bhapkar, **Pierre Kawak**, and David S. Simmons. "Insights into the Dependence of Elastomeric Nanocomposite Mechanics on Nanoparticulate Properties". Annual Meeting. American Institute of Chemical Engineers. Orlando, FL, 2023.
- [14] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Dissecting the Payne Effect: How Filler-Polymer Competition Reinforces Elastomeric Nanocomposites". IOP Polymer Physics Group Graduate Symposium. Institute of Physics. Virtual, 2023.
- [13] **Pierre Kawak**. "Career Paths in Physics". Physics Colloquia Series. University of South Florida Department of Physics. Tampa, FL, 2023.
- [12] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Exploring Mechanisms of Enhanced Dissipation in Nanoparticle-filled Rubber Using Molecular Dynamics". Annual Postdoctoral Research Symposium. University of South Florida. Tampa, FL, 2023.
- [11] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Exploring mechanisms of enhanced dissipation in nanoparticle-filled rubber using molecular dynamics". March Meeting. American Physical Society. Las Vegas, NV, 2023.
- [10] Douglas R. Tree and **Pierre Kawak**. "Free Energy Analysis of Crystal Nucleation of Semiflexible Polymers". March Meeting. American Physical Society. Las Vegas, NV, 2023.
- [9] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Spatially resolving energy dissipation in molecular dynamics of polymer nanocomposites". March Meeting. American Physical Society. Las Vegas, NV, 2023.
- [8] **Pierre Kawak**, Dakota S. Banks, and Douglas R. Tree. "Acute Sensitivity of Polymer Crystallization Phase Behavior to Intermolecular Interactions". Annual Meeting. American Institute of Chemical Engineers. Phoenix, AZ, 2022.
- [7] **Pierre Kawak**, Dakota S. Banks, and Douglas R. Tree. "Free Energy Analysis of Polymer Crystal Nucleation Indicates Cooperative Crystallization and Nematic Alignment". March Meeting. American Physical Society. Chicago, IL, 2022.
- [6] **Pierre Kawak**, Dakota S. Banks, and Douglas R. Tree. "Free Energy Surfaces for Homogeneous Nucleation in a Polymer Melt". Annual Meeting. American Institute of Chemical Engineers. Boston, MA, 2021.
- [5] **Pierre Kawak**, Dakota S. Banks, and Douglas R. Tree. "GPU-accelerated Wang-Landau Simulation of Polymer Crystallization". March Meeting. American Physical Society. Virtual, 2021.
- [4] **Pierre Kawak**, Andrew S. Gibson, Logan S. Brown, Beverly Delgado, Douglas R. Tree, and Dakota S. Banks. "Investigating Primary Nucleation in Polymer Melts using GPU-Accelerated Wang-Landau Simulations". Annual Meeting. American Institute of Chemical Engineers. Virtual, 2020.
- [3] **Pierre Kawak**, Andrew S. Gibson, Logan S. Brown, Beverly Delgado, and Douglas R. Tree. "Wang-Landau Simulation of the Free Energy Surface of Crystallization in a Polymer Melt". March Meeting. American Physical Society. Virtual, 2020.
- [2] **Pierre Kawak**, Vinod Paul, Paul Kawak, Rita Kassermally, Fatme Lahib, Rute F. Vitor, Mohammad Al-Sayah, and Ghaleb A. Hussein. "Doxorubicin-Encapsulated, Estrone-Appended Liposomes Triggered by Ultrasound for the Treatment of Breast Cancer". Graduate Students Research Conference. UAE Ministry of Education. Khalifa University, Abu Dhabi, UAE, 2017.
- [1] **Pierre Kawak**, Christian C. Momah, Mohamed A. Elkhodiry, Shaima R. Suwaidi, Dina Gadalla, Fatehia M. Banamah, Rute F. Vitor and Hesham G. Moussa, Ana M. Martins and Mohammad Al-Sayah, and Ghaleb A. Hussein. "A Peptide-Targeted Nanodelivery System Triggered by Ultrasound for Anti-cancer Therapy". Life Sciences Grand Challenges Conference. Institute of Engineering and Electronics Engineering. Khalifa University, Abu Dhabi, UAE, 2016.